Final Project Report

Project Direction Changes

Throughout the implementation of the project, the team has adhered to the project's initial direction, which is to offer a data-driven learning platform to provide insightful feedback to AcademicAlly users (Students and Faculty). The team has enriched the functionalities by collecting relevant data and visualizing them through an interactive dashboard.

Application Usefulness

The application's implementation has ensured its utility to provide the core functionalities (routine application usage by students and faculties) and metrics collection through filters, aggregations, and advanced queries. The core functionalities and the metrics satisfy the application's usefulness, as committed to in the initial proposal.

The application's users can make wise decisions based on the feedback and metrics delivered by the application. We are yet to cover the plagiarism checker functionality as it should be highly customizable in nature. The framework of the plagiarism checker should provide multiple parameters for the faculty to decide the nature and extent of checks, and this task of determining the customizability of the checker needs more analysis and feedback from faculty.

Data/Schema Changes

The sources of data for the application have remained the same throughout the project.

The schema has been augmented with a new table 'IncorrectResponses' to persist the incorrect assignment's responses in the database. We could have evaluated the incorrect responses for every assignment, but persisting them in the DB to derive useful metrics using this data made more sense. 'IncorrectResponses' is a weak entity with the 'Submission' table as the identifying entity.

Relational schema for the newly added IncorrectResponses table:-

```
CREATE TABLE IncorrectResponses(
        ResponseID INT NOT NULL AUTO_INCREMENT,
```

```
        SubmissionID VARCHAR (20),
        Topic VARCHAR (255),
        PRIMARY KEY(ResponseID,SubmissionID),
        FOREIGN KEY(SubmissionID) REFERENCES Submission(SubmissionID)
);
```

Table implementation/ER diagram changes

As explained in the previous section, the ER diagram is augmented with a new weak entity “IncorrectResponses” with the ‘Submission’ table as its identifying entity.

This is used primarily to store the topics of multiple-choice questions incorrectly answered by the student.

The newer design is more suitable for AcademicAlly as, with the help of this new table, we can add the ‘recommended topics’ feature that provides the students with the topics they are weakest in. Incorrect responses could be computed on the fly; however, it made more sense to persist this data as it can be used to provide tailor-made suggested topics.

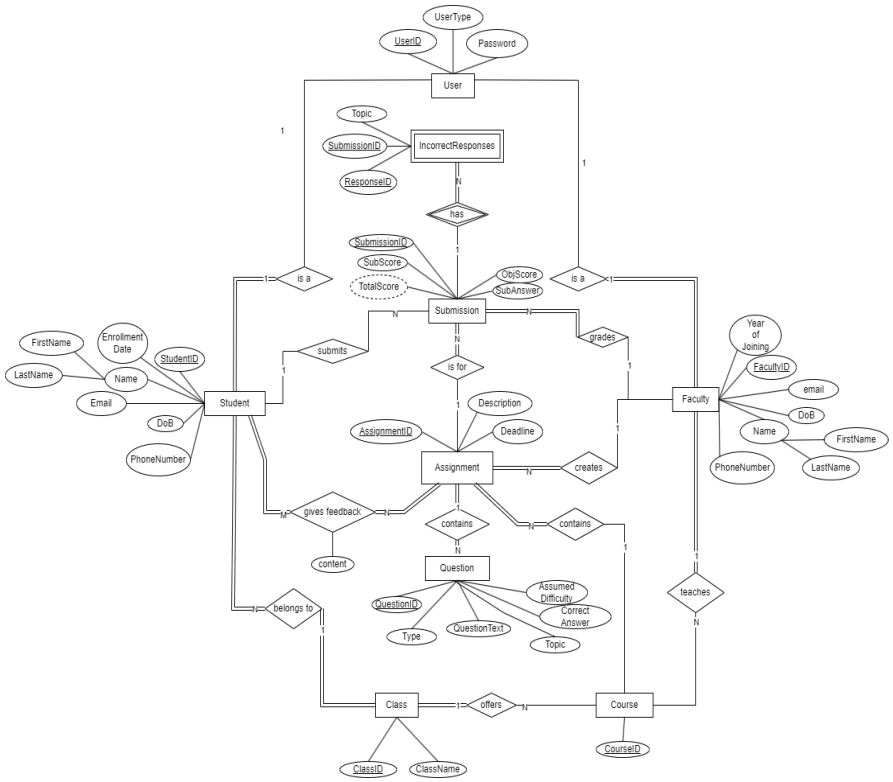


Fig 1: IncorrectResponses added to ER Diagram

Functionality additions/removals

Functionality Added:-

a. Deadline viewer for students -

In the student dashboard, we added a table to view the upcoming deadlines for the assignments yet to be submitted. This view would make students aware of approaching deadlines and help students in prioritizing tasks.

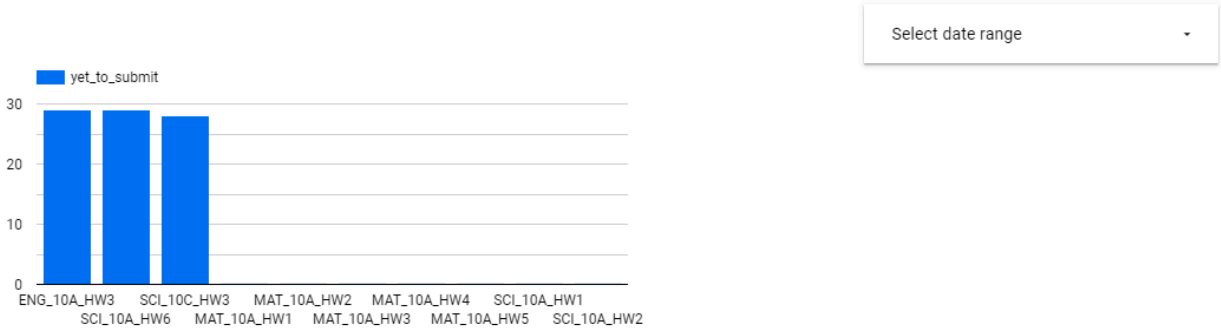
Upcoming Deadlines

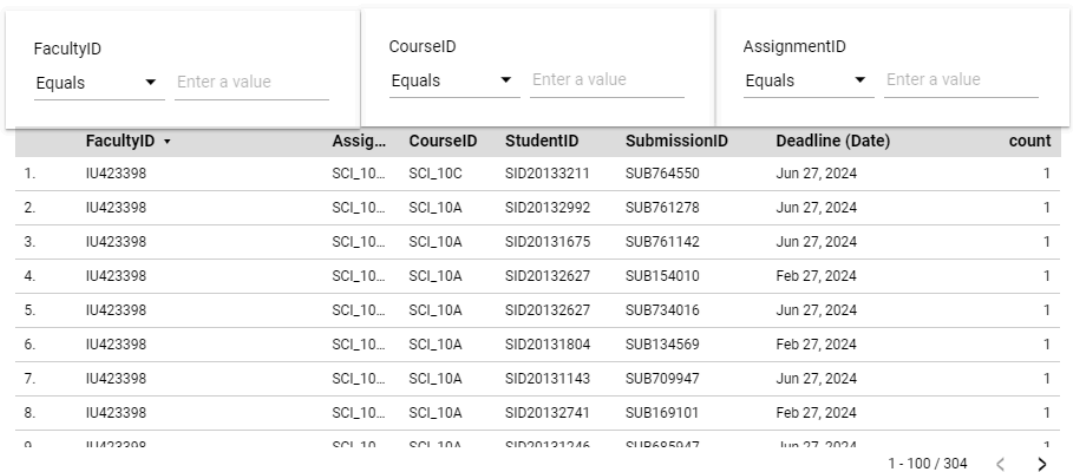
ASSIGNMENT ID	UPCOMING DEADLINE
SCI_10A_HW7	Fri, 31 May 2024 00:00:00 GMT
GK_10A_HW3	Thu, 27 Jun 2024 00:00:00 GMT
ENG_10A_HW4	Tue, 27 Aug 2024 00:00:00 GMT
GK_10A_HW4	Tue, 27 Aug 2024 00:00:00 GMT
ENG_10A_HW5	Sun, 27 Oct 2024 00:00:00 GMT
GK_10A_HW5	Sun, 27 Oct 2024 00:00:00 GMT

Fig 2: Upcoming assignment deadlines of a student in AcademicAlly database

b. Live assignment submission tracker-

For faculties, we have constructed a bar plot that displays the number of pending submissions for each assignment. This plot is interactive, where the faculty can select the course, and assignmentID from the filters and visualize the plot. Conversely, an assignment from the plot can be selected to visualize the submission details for the chosen assignment in the table. We have leveraged Google Looker Studio to deliver this functionality.



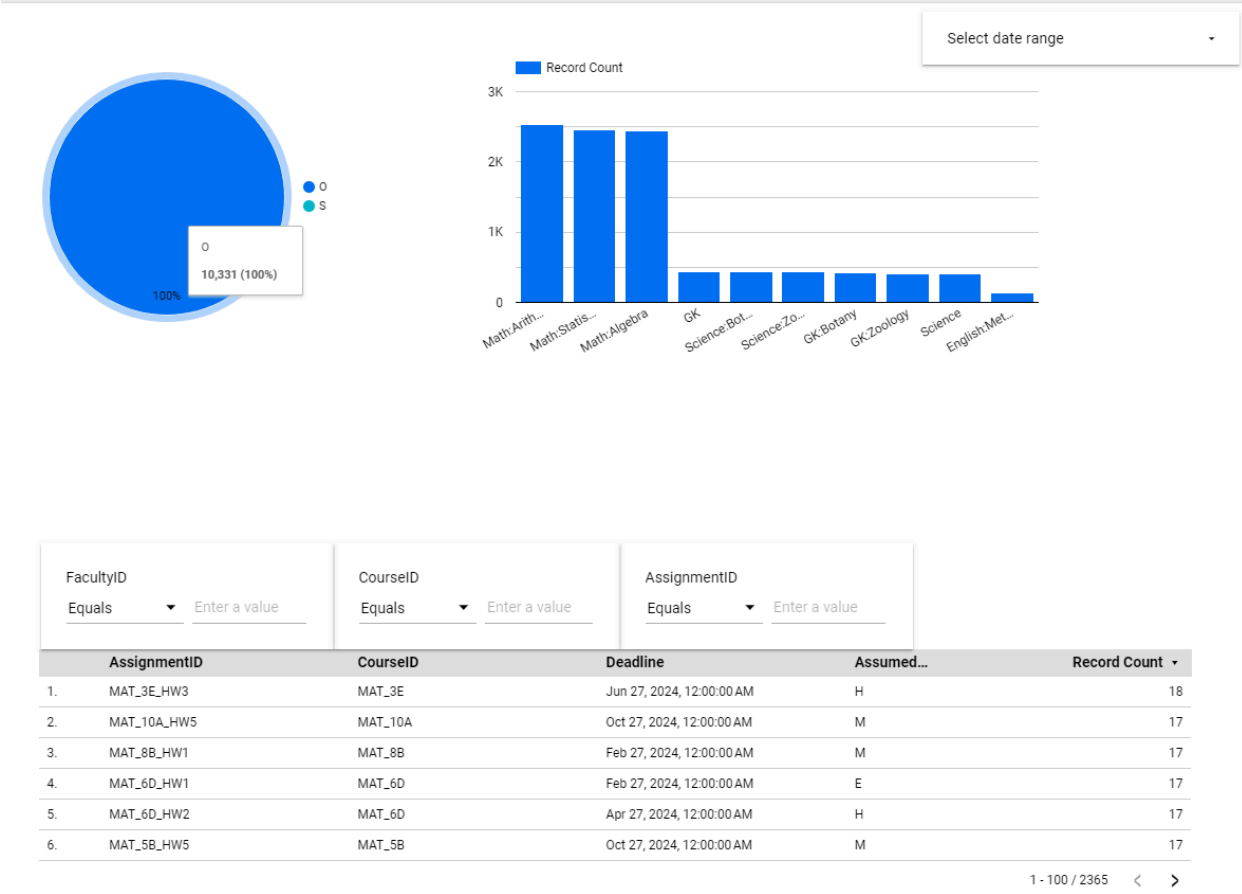


	FacultyID	Assig...	CourseID	StudentID	SubmissionID	Deadline (Date)	count
1.	IU423398	SCI_10...	SCI_10C	SID20133211	SUB764550	Jun 27, 2024	1
2.	IU423398	SCI_10...	SCI_10A	SID20132992	SUB761278	Jun 27, 2024	1
3.	IU423398	SCI_10...	SCI_10A	SID20131675	SUB761142	Jun 27, 2024	1
4.	IU423398	SCI_10...	SCI_10A	SID20132627	SUB154010	Feb 27, 2024	1
5.	IU423398	SCI_10...	SCI_10A	SID20132627	SUB734016	Jun 27, 2024	1
6.	IU423398	SCI_10...	SCI_10A	SID20131804	SUB134569	Feb 27, 2024	1
7.	IU423398	SCI_10...	SCI_10A	SID20131143	SUB709947	Jun 27, 2024	1
8.	IU423398	SCI_10...	SCI_10A	SID20132741	SUB169101	Feb 27, 2024	1

Fig 3: assignment submission tracker for the faculty

c. Topic tracker for faculty -

For the faculty, we have created a bar plot and a pie chart to visualize the topics covered and the distribution of subjective to objective questions respectively. These plots can be viewed at the course and assignment levels (by selecting the assignmentID in the filter). They help the faculty determine the topics and the number of questions asked for each topic. This metric can be used to tune the question distribution in assignments by increasing the frequency of questions from important topics.



	AssignmentID	CourseID	Deadline	Assumed...	Record Count
1.	MAT_3E_HW3	MAT_3E	Jun 27, 2024, 12:00:00 AM	H	18
2.	MAT_10A_HW5	MAT_10A	Oct 27, 2024, 12:00:00 AM	M	17
3.	MAT_8B_HW1	MAT_8B	Feb 27, 2024, 12:00:00 AM	M	17
4.	MAT_6D_HW1	MAT_6D	Feb 27, 2024, 12:00:00 AM	E	17
5.	MAT_6D_HW2	MAT_6D	Apr 27, 2024, 12:00:00 AM	H	17
6.	MAT_5B_HW5	MAT_5B	Oct 27, 2024, 12:00:00 AM	M	17

Fig 4: Question Topic tracker for the faculty

d. Average Difficulty vs Average marks for assignments -

We identified an important metric that can help faculty assess the difficulty of the assignments by observing the student's performance. While creating assignments, the faculty assumes the difficulty level of each question. Looking at the student's performance in assignments, the faculty can assess whether the assumption of the difficulty level of the assignment (averaged difficulty of all questions in the assignment) was correct, and if the student's performance deviates from the assumption, then the faculty can take appropriate action by tuning the difficulty of questions or modifying the teaching methods for topics covered under those assignments.

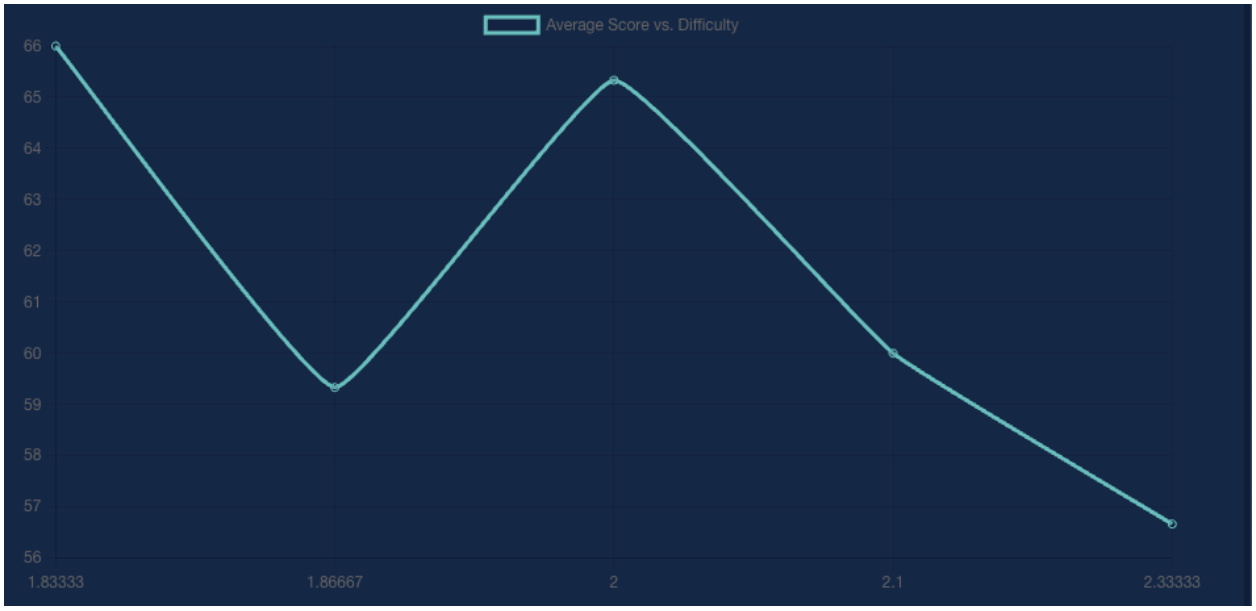


Fig 5: Average difficulty (x-axis) vs Average score (y-axis) plot

Functionalities Descoped:-

a. Plagiarism Checker -

   Initially, we planned to implement a plagiarism checker to check subjective answers. However, through deeper analysis, we found that the framework of the checker should provide multiple parameters for the faculty to decide the nature and extent of the checks, and this task of determining the customizability of the checker needs more analysis and feedback from the faculty.

b. Extracurricular activity tracker -

   This tracker aimed to establish a relationship between student participation in extracurricular activities and academic performance. However, the diversity of extracurricular activities mandates a deeper analysis of the exact metrics that could contribute to establishing the desired relationship.

Advanced Database Queries Contribution

We leveraged advanced queries with joins, subqueries, and aggregations to reveal embedded knowledge from the database.

a. Adv query 1 - Finding the average difficulty of the assignment/course.

   i. With the average difficulty score of each assignment, the faculty can tailor his assignments to provide a steadily increasing difficulty curve as the semester progresses.

   ii. The average difficulty score can be used to analyze the class's performance relative to the assignment difficulty. This analysis can suggest a change in teaching method or assignment difficulty if required.

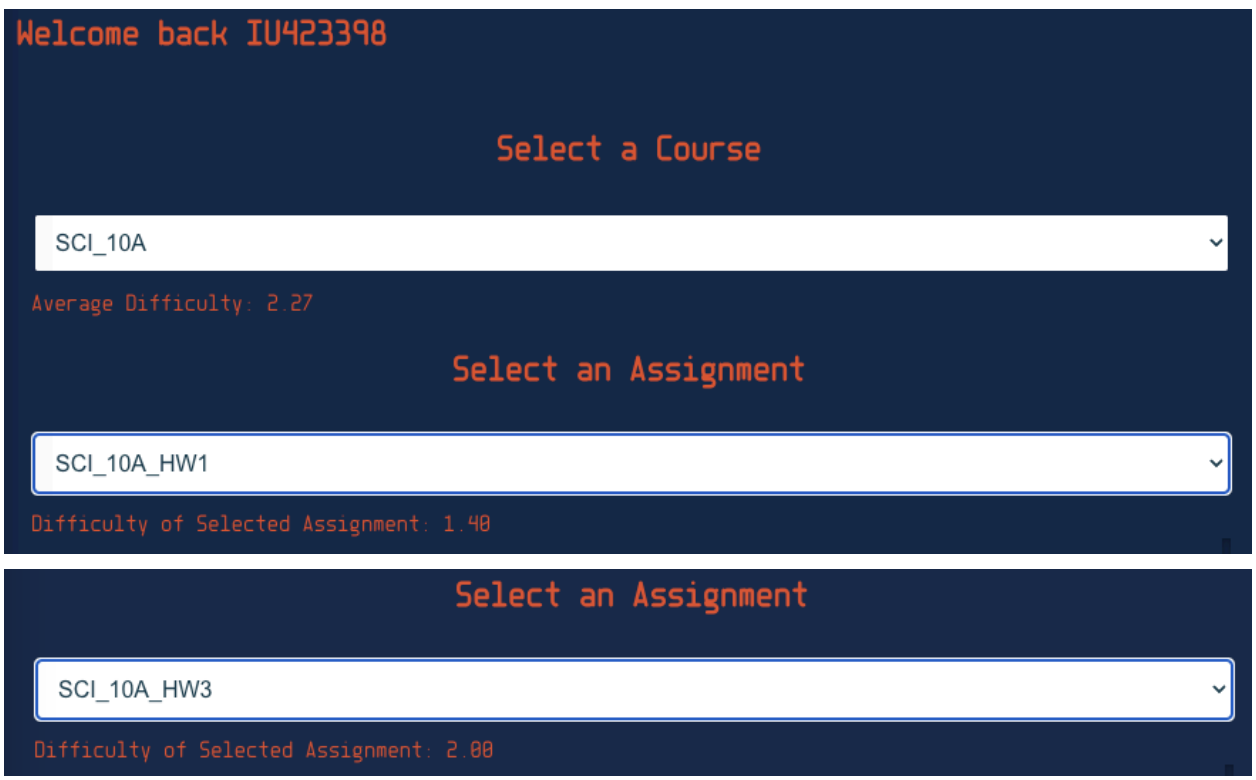


Fig 6: UI showing the assumed difficulty of assignments and aggregated difficulty for the course

b. Adv query 2 - Consolidated upcoming deadline viewer

   i. AcademicAlly displays all the upcoming and unsubmitted assignment deadlines to students on their main dashboard for all the courses they have undertaken.

   ii. By clearly seeing upcoming deadlines, students can better plan their study schedules and allocate time accordingly, thus improving their work quality and overall academic performance. *Please refer to Figure 2.*

c. Adv query 3 - Weak students list for faculty

   i. AcademicAlly displays the details of all students considered 'weak' in a course taught by the faculty. The students are considered 'weak' if they have scored less than the class average in two or more assignments.

   ii. AcademicAlly also displays the assignments for which they have gotten lower scores thus helping the faculty provide additional assistance for those particular assignments.

Weak Students

STUDENT ID	NUMBER OF ASSIGNMENTS	LIST OF ASSIGNMENTS
SID20131231	3	SCI_10A_HW3, SCI_10A_HW2, SCI_10A_HW5
SID20131384	4	SCI_10A_HW3, SCI_10A_HW1, SCI_10A_HW5, SCI_10A_HW4
SID20131529	3	SCI_10A_HW3, SCI_10A_HW1, SCI_10A_HW5
SID20131582	4	SCI_10A_HW3, SCI_10A_HW1, SCI_10A_HW5, SCI_10A_HW4
SID20131703	3	SCI_10A_HW3, SCI_10A_HW1, SCI_10A_HW5

Fig 7: List of weak students for a particular course taught by the faculty

d. Adv query 4 - Average Score and Submitted Responses statistics

   i. AcademicAlly presents the course statistics(namely, the average score of each assignment and the number of submitted responses for the assignment) for each course the faculty teaches. *Please refer to Figure 5.*

Technical Challenges

a. Interactive Dashboard for visualization
   - We leveraged the chart.js library to create customized graphical representations for students and faculty in their respective dashboards. Google Looker Studio was also integrated to achieve greater interactivity.

b. Temporary tables in Stored Procedure
   - Using temporary tables before the stored procedure call gave us more flexibility on the information we can pass to the stored procedure apart from the regular in and out arguments. However, attention must be paid to ensure the tables are correctly cleaned after usage.

c. Finding the sweet spot between the number of endpoints and the complexity of the endpoint
   - Finding the tradeoff between the number of endpoints and the complexity enhances API usability and maintainability. This ensured that endpoints were neither too granular nor overloaded with different features, allowing easier integration with the React Frontend.

d. Defining ingress and egress rules for application access
   - Establishing clear ingress and egress rules ensures that only legitimate traffic can enter or leave the system, preventing unauthorized access. Following the rule of least privilege, we implemented appropriate IAM role bindings for the application service accounts, thus maintaining a clear hierarchy structure.

Other Project Changes

In comparing the final project with the original proposal, there have been no significant changes to the project's intended aim of providing a data-driven learning platform. However, significant augmentations with respect to visualizations were done by the

team to derive essential metrics. This further improved the productivity of the platform's users by enabling them to make informed decisions. The detailed descriptions of these augmentations can be found in Section 5 of this document.

Future Work

Some future works that can be undertaken to improve the application are:

- RAG implementation

  Retrieval Augmented Generation can be implemented in the Generative AI feedback feature to improve the relevancy and usefulness of the responses from the Gemini API.

- Collaborative Chat Interface among students
  - As of now, the app focuses on observing the student's performance and their feedback to derive optimal teaching methods. However, a social network among students can be established through an open chat server interface whose data can be leveraged to observe student collaboration and how it can contribute to tuning the teaching methods.
  - This chat interface can also assist in adding a group projects feature in AcademicAlly.

Division of Labor

Common responsibilities of the team were:

- Problem Identification
- Schema Design
- Advanced Queries
- Tech Stack choice
- Demo use case creation
- Procedure and Triggers workflow design

Individual responsibilities:-

1. Vaibhav Bhandari:-
   a. Backend development involving DB client implementation for student workflow.
   b. Developed backend responsiveness to front-end calls.
   c. Front-end development for the assignment submission workflow.

2. Bhuvanesh Shukla
    a. Front-to-backend integration through REST client development for front-end-to-backend calls.
    b. Creation of Google Data Studio with metrics for the application.
    c. Backend development for populating the tables with data and faculty metrics computation.

3. Aditya jhunhunwala
    a. Backend development involving Procedure development.
    b. Frontend development involving the creation of Deadline and Student feedback functionality.
    c. Backend development involving DB client implementation for faculty workflow.

4. Gautam Kher
    a. Backend development involving Trigger development.
    b. Frontend development involving graph and pie chart plots for student dashboard.
    c. Backend development for populating the tables with data and student metrics computation.